

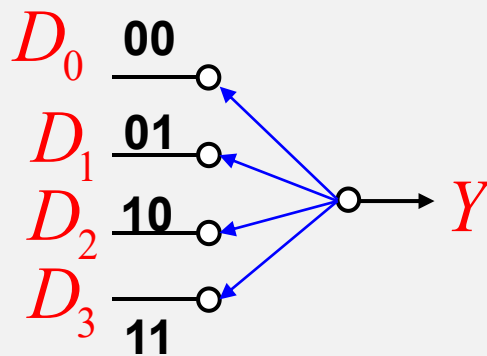
第四章 组合逻辑电路设计(三)

秦磊华 计算机学院

4.7 多路选择器 (Multiplexer)设计

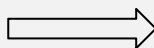
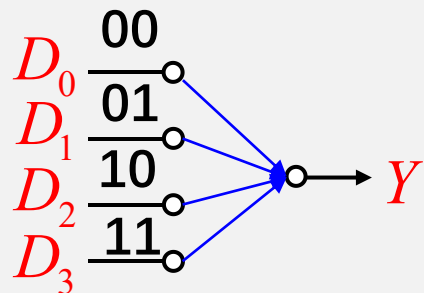
1. 多路选择器的基本功能

从一组输入数据中，选择出某一个数据，完成这种功能的逻辑电路称为数据选择器（或称为多路选择开关）



4.7 多路选择器(Multiplexer)设计

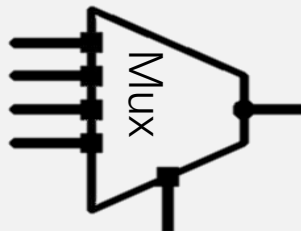
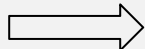
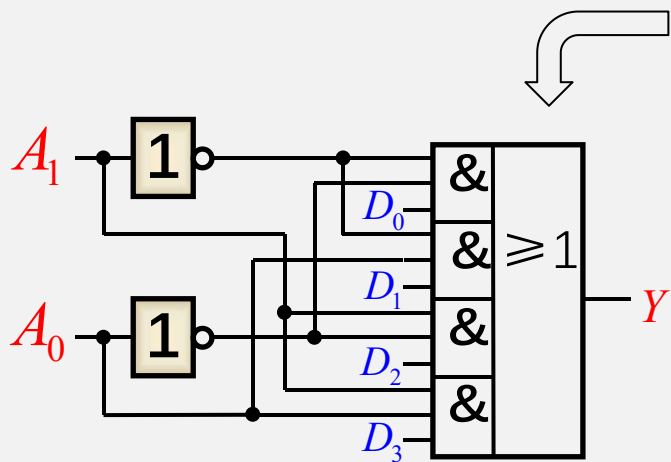
2. 4路数据选择器的设计 (MUX)



D	A ₁	A ₀	Y
D ₀	0	0	D ₀
D ₁	0	1	D ₁
D ₂	1	0	D ₂
D ₃	1	1	D ₃

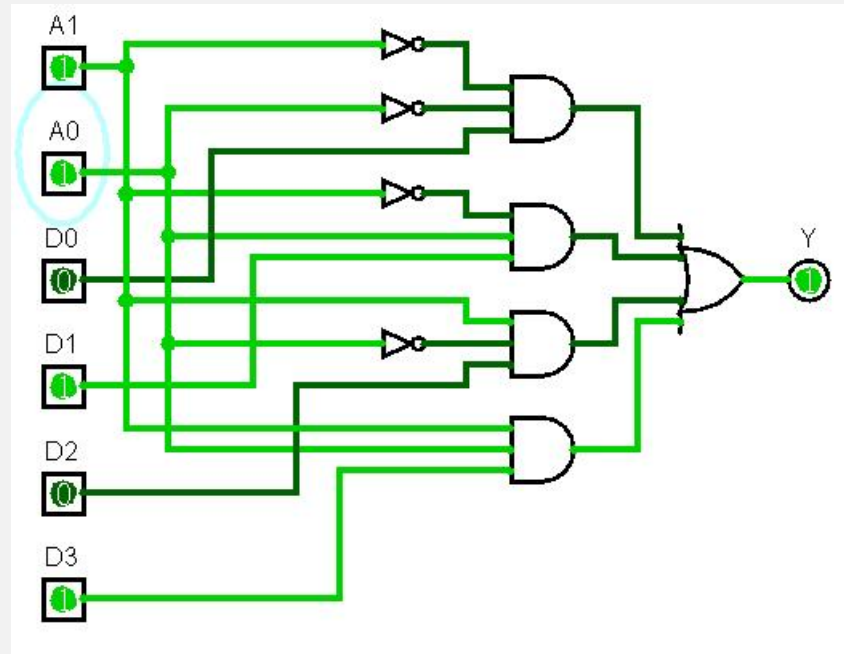


$$Y = \overline{A_1} \overline{A_0} D_0 + \overline{A_1} A_0 D_1 + A_1 \overline{A_0} D_2 + A_1 A_0 D_3$$



4.7 多路选择器(Multiplexer)设计

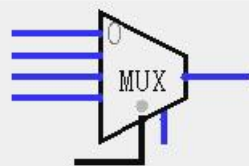
$$Y = \bar{A}_1 \bar{A}_0 D_0 + \bar{A}_1 A_0 D_1 + A_1 \bar{A}_0 D_2 + A_1 A_0 D_3$$



4.7 多路选择器(Multiplexer)设计

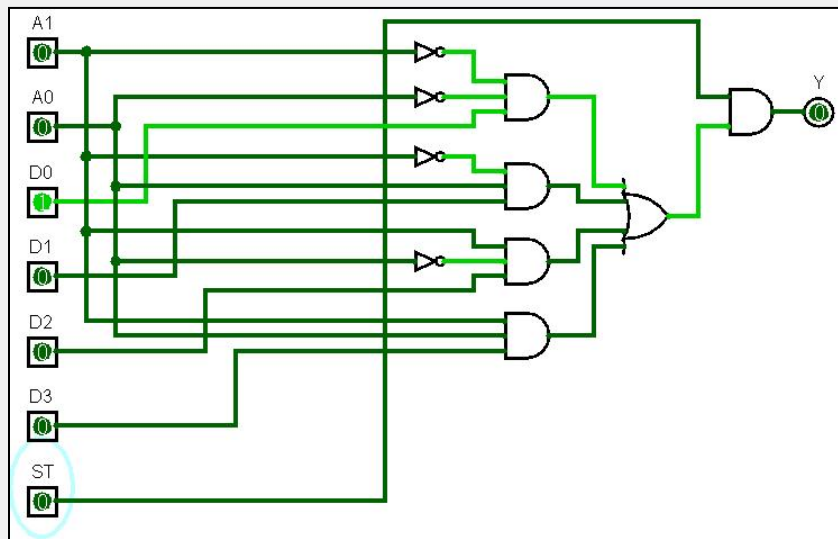
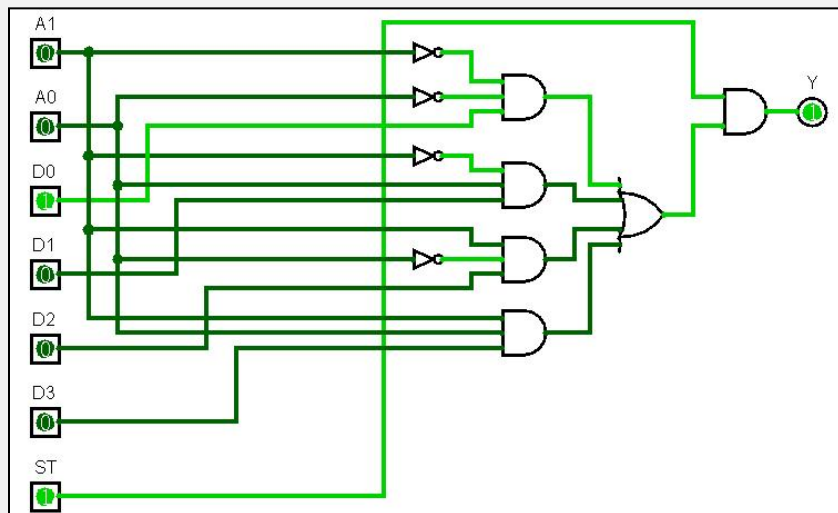
$\overline{ST}_1$	A_1	A_0	Y_1
1	X	X	0
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3

$$Y = \overline{\overline{ST}_1} (\overline{A_1} \overline{A_0} D_0 + \overline{A_1} A_0 D_1 + A_1 \overline{A_0} D_2 + A_1 A_0 D_3)$$



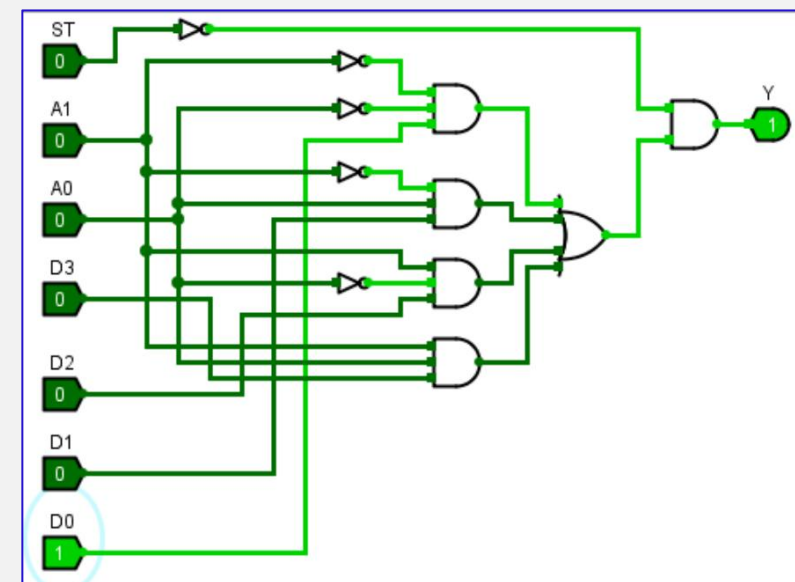
Enable: when not 0, output is the selected input

4.7 多路选择器(Multiplexer)设计



使能信号为正

$\overline{ST}_1$	A_1	A_0	Y_1
1	X	X	0
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3



使能信号为负

4.7 多路选择器(Multiplexer)设计

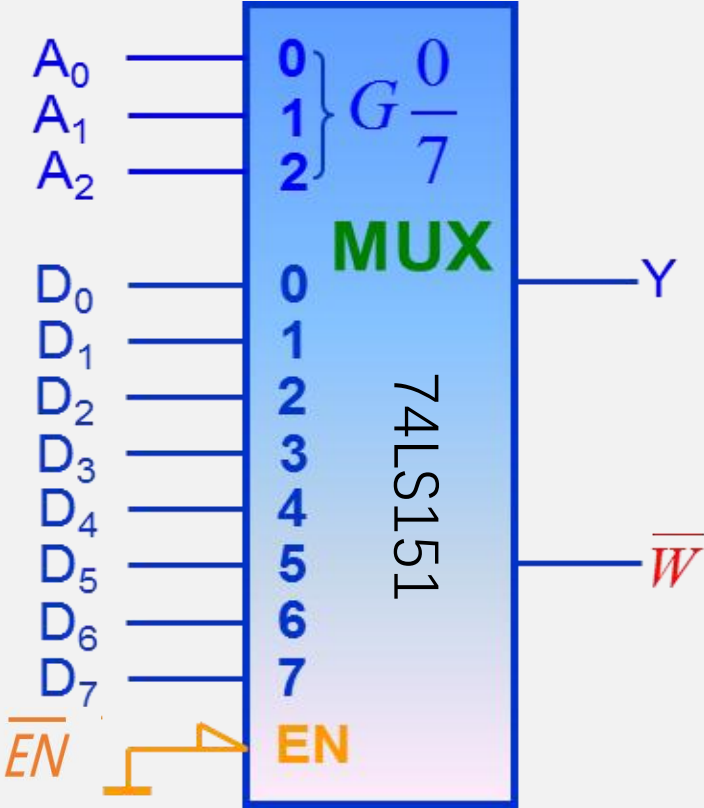
3.带使能和可扩展功能的8路数据选择器设计

$\overline{EN}$	A_2	1	0		—
1	X	X	X	0	1
0	0	0	0	D_0	$\overline{D_0}$
0	0	0	1	D_1	$\overline{D_1}$
0	0	1	0	D_2	$\overline{D_2}$
0	0	1	1	D_3	$\overline{D_3}$
0	1	0	0	D_4	$\overline{D_4}$
0	1	1	1	D_5	$\overline{D_5}$
0	1	0	0	D_6	$\overline{D_6}$
0	1	1	1	D_7	$\overline{D_7}$

$\overline{EN}$: 选通端，低有效。

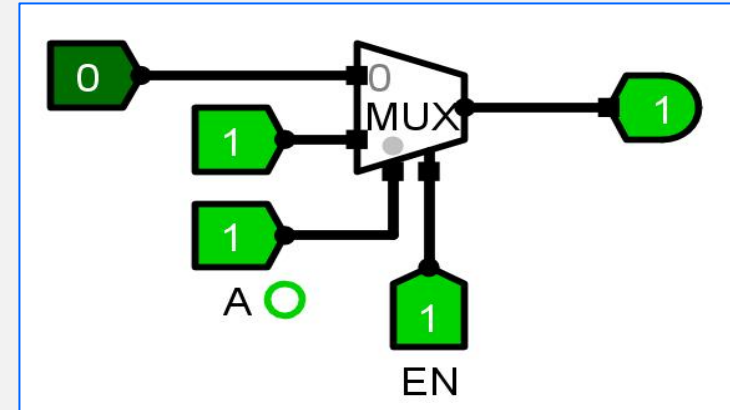
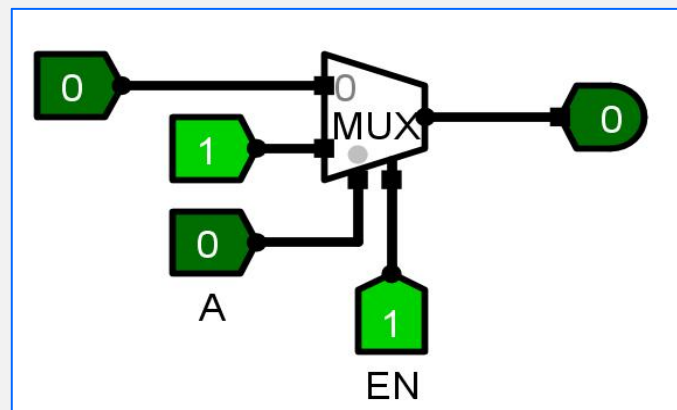
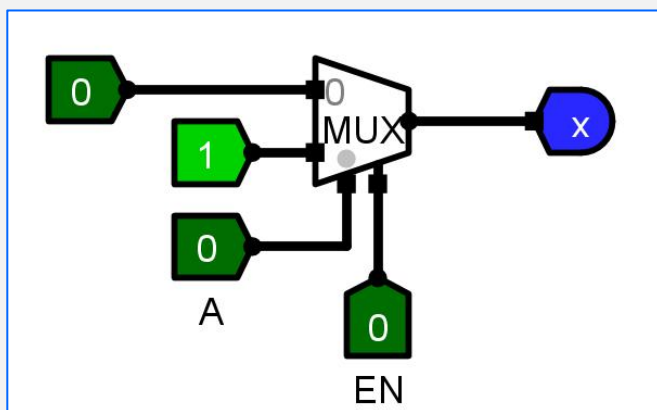
$Y, \overline{W}$: 互补输出端。

可参照4路选择器写出Y逻辑表达式



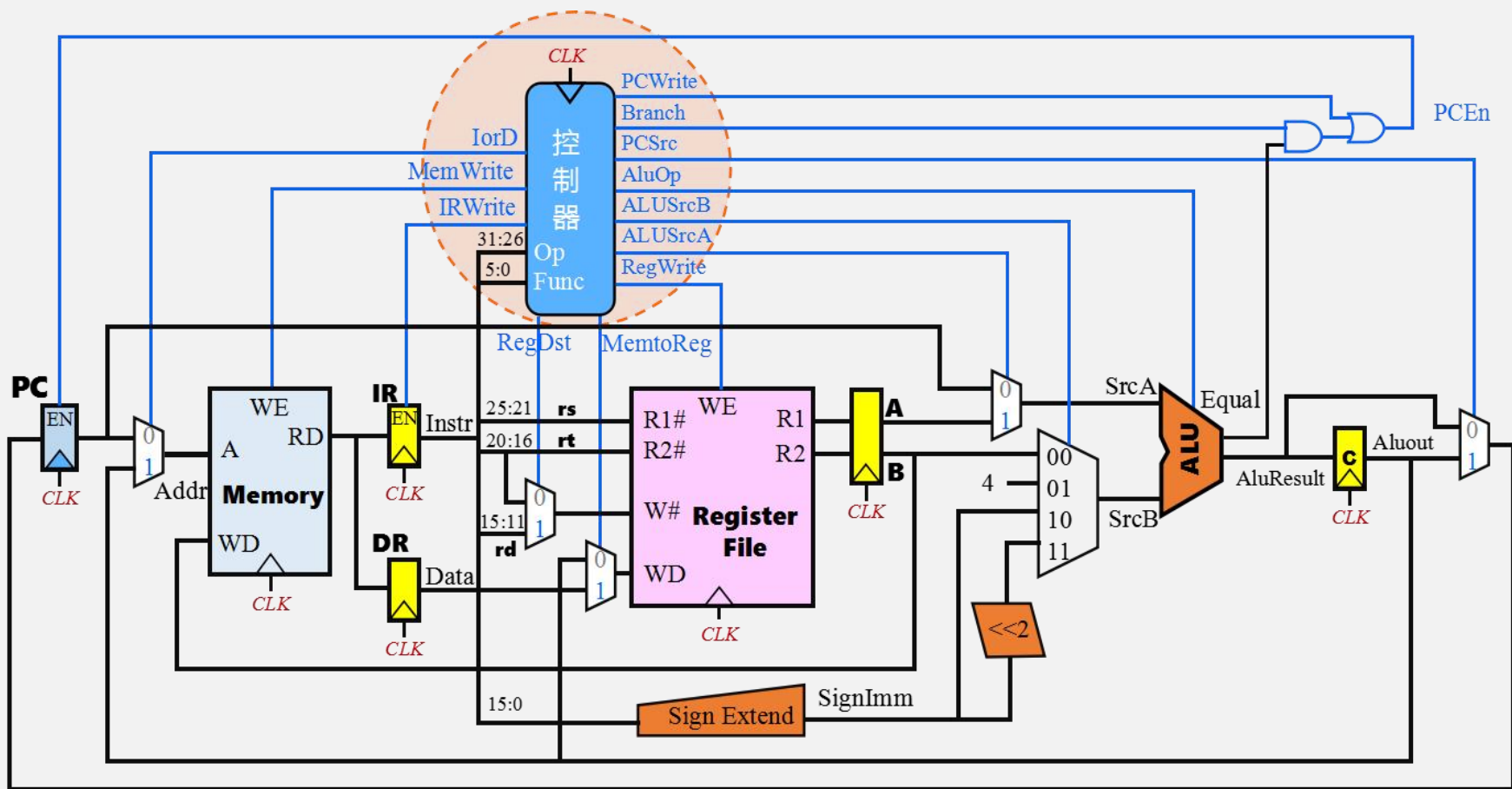
$$Y = (\overline{A_2}\overline{A_1}\overline{A_0} D_0 + \overline{A_2}\overline{A_1}A_0 D_1 + \overline{A_2}A_1\overline{A_0} D_2 + \overline{A_2}A_1A_0 D_3 + A_2\overline{A_1}\overline{A_0} D_4 + A_2\overline{A_1}A_0 D_5 + A_2A_1\overline{A_0} D_6 + A_2A_1A_0 D_7) \overline{EN}$$

4.7 多路选择器(Multiplexer)设计



4.7 多路选择器(Multiplexer)设计

4.数据选择器的应用

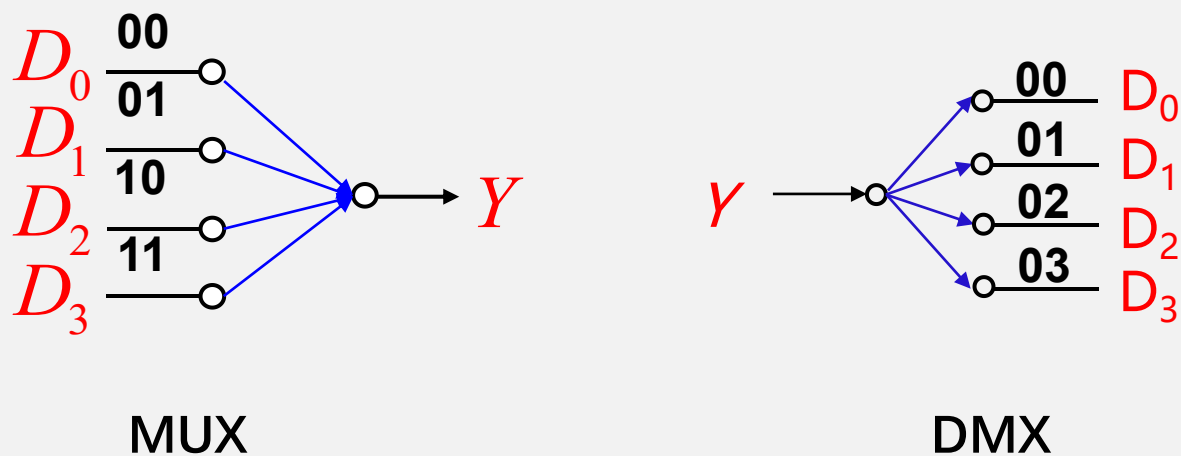


同一目标有多个数据来源时，在其入口处需使用多路选择器

4.8 多路分配器(解复用器 Demultiplexer)

1. 多路分配器的基本功能

将1个输入数据，根据需要传送到m个输出端的任何一个输出端的电路，称为数据分配器、多路分配器或解复用器，其逻辑功能正好与多路选择器相反。



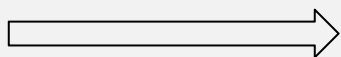
4.8 多路分配器(解复用器 Demultiplexer)

2. 多路分配器的设计

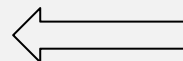
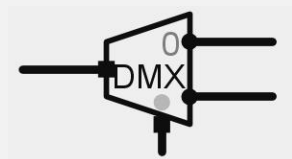
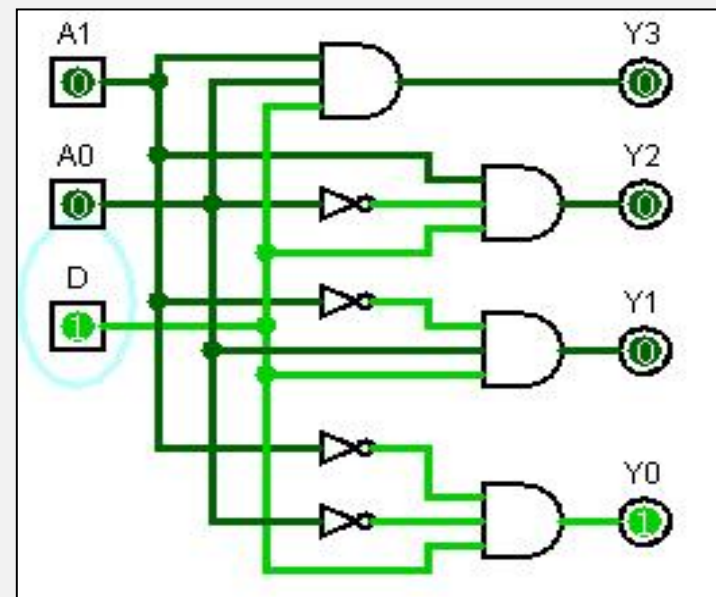
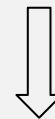
A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	D
0	1	0	0	D	0
1	0	0	D	0	0
1	1	D	0	0	0

A_1	A_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

多路选择器真值表



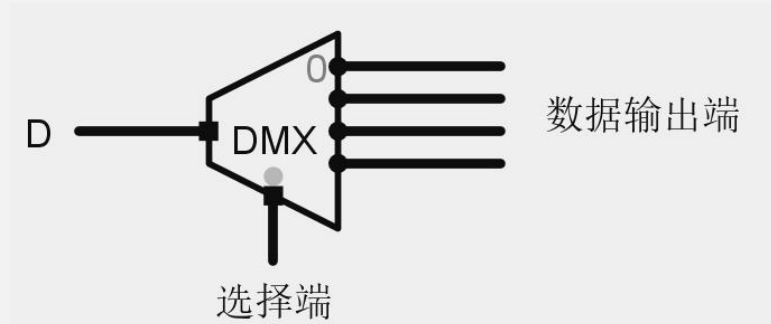
$$\begin{aligned} Y_0 &= \overline{A_1} \overline{A_0} D & Y_1 &= \overline{A_1} A_0 D \\ Y_2 &= A_1 \overline{A_0} D & Y_3 &= A_1 A_0 D \end{aligned}$$



4.9 多路选择器、多路分配器、译码器比较



$$Y = \overline{A_1}\overline{A_0}D_0 + \overline{A_1}A_0D_1 + A_1\overline{A_0}D_2 + A_1A_0D_3$$



$$Y_0 = \overline{A_1}\overline{A_0}D \quad Y_1 = \overline{A_1}A_0D \quad Y_2 = A_1\overline{A_0}D \quad Y_3 = A_1A_0D$$



$$Y_0 = \overline{A_1}\overline{A_0} \quad Y_1 = \overline{A_1}A_0 \quad Y_2 = A_1\overline{A_0} \quad Y_3 = A_1A_0$$

4.10 基于基本组合逻辑功能部件的组合逻辑设计

1. 利用变量译码器实现组合逻辑函数

A_1	A_0	Y3	Y2	Y1	Y0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

$$Y_3 = A_1 A_0 \quad Y_2 = A_1 \bar{A}_0$$

$$Y_1 = \bar{A}_1 A_0 \quad Y_0 = \bar{A}_1 \bar{A}_0$$

一个n变量输入的变量译码器，其输出包含了n个输入变量的全部最小项。用n变量译码器加输出门就能实现任何形式的输入变量不大于n的组合逻辑函数。

4.10 基于基本组合逻辑功能部件的组合逻辑设计

例1 用译码器实现一组多输出函数

$$F_1 = \overline{A}\overline{B} + \overline{B}C + AC$$

$$F_2 = \overline{A}\overline{B} + B\overline{C} + ABC$$

$$F_3 = \overline{A}C + BC + A\overline{C}$$

解：三输入变量的多输出函数，用3-8译码器实现

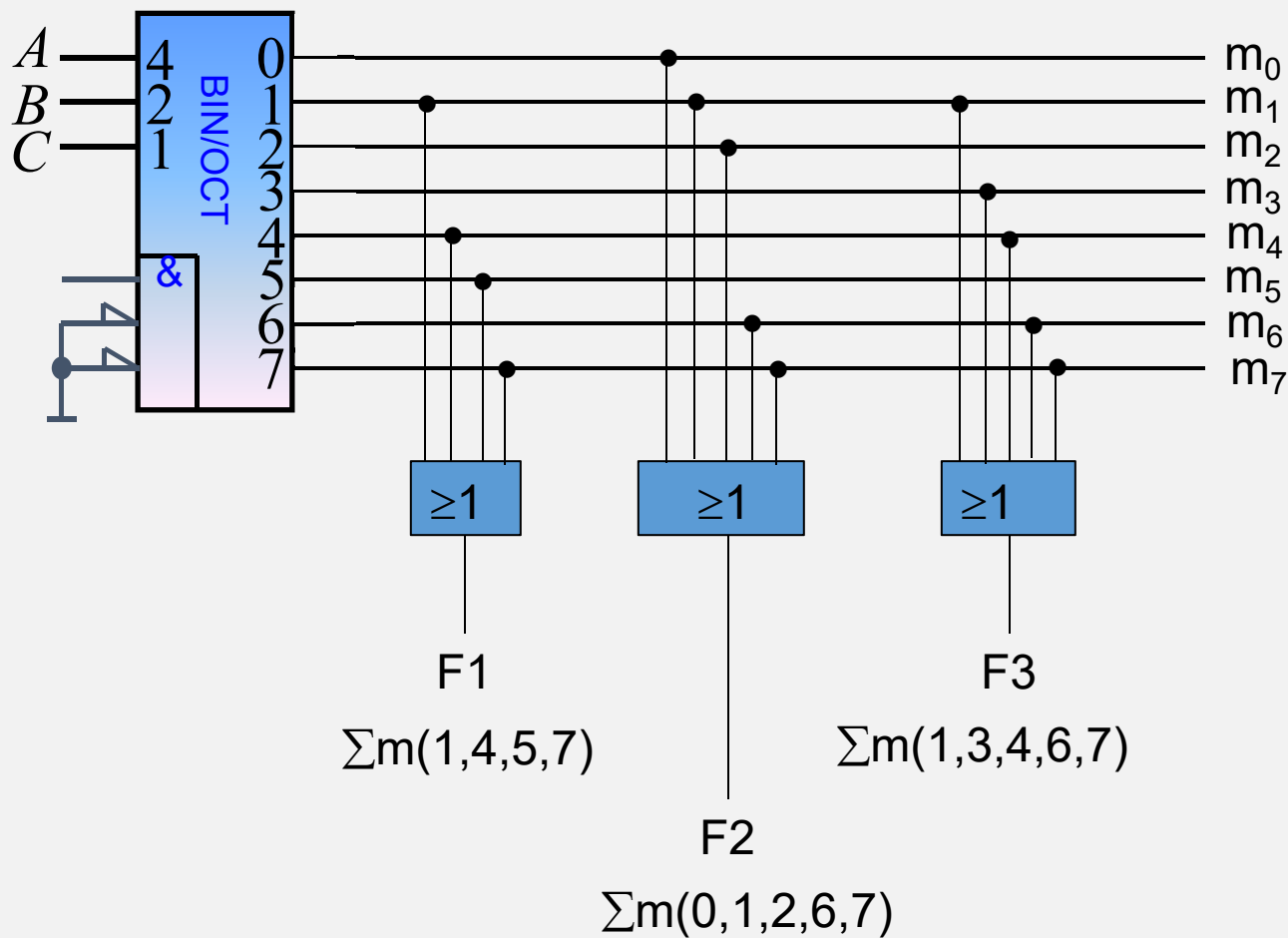
将多输出函数写成最小项之和形式,再配合适当的逻辑门即可。

$$F_1 = \overline{A}\overline{B} + \overline{B}C + AC = \sum m(1,4,5,7)$$

$$F_2 = \overline{A}\overline{B} + B\overline{C} + ABC = \sum m(0,1,2,6,7)$$

$$F_3 = \overline{A}C + BC + A\overline{C} = \sum m(1,3,4,6,7)$$

4.10 基于基本组合逻辑功能部件的组合逻辑设计



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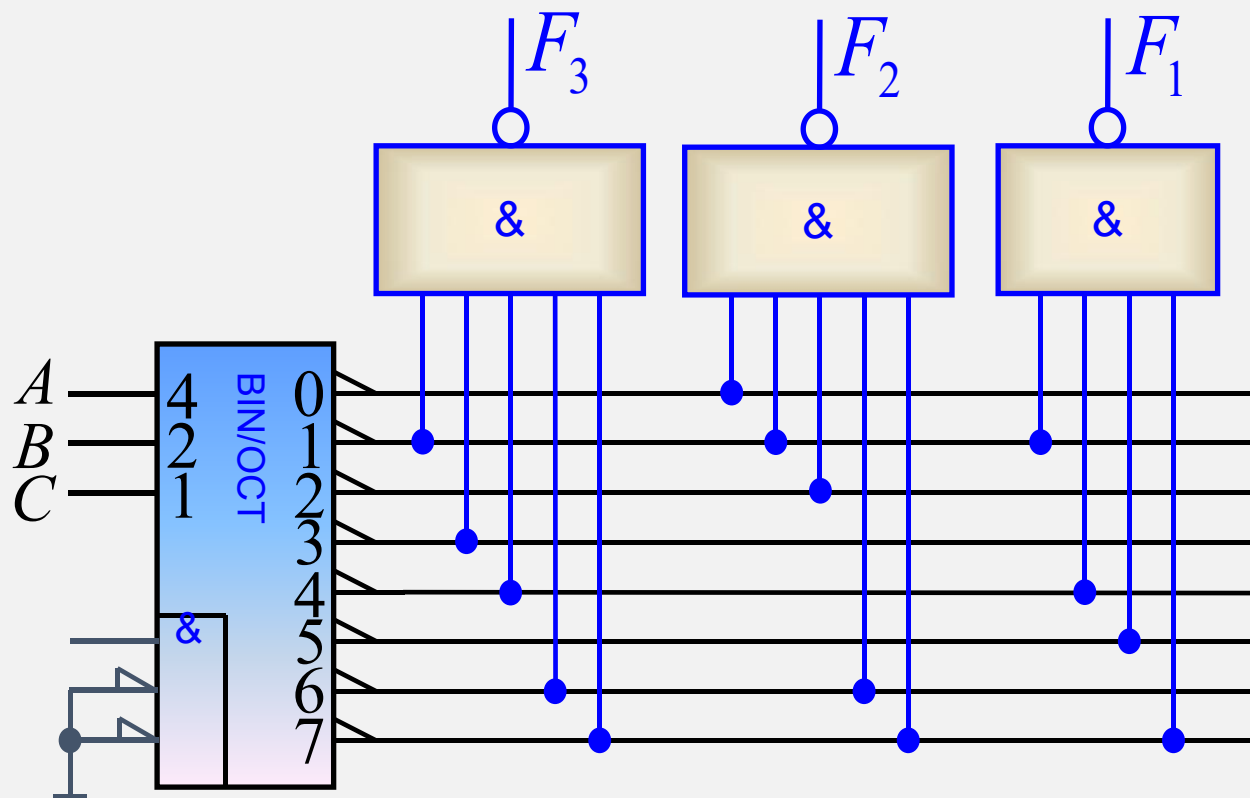
若译码器是以反变量形式输出，即输出的是 $\overline{m}_i$ ，则：

$$\begin{aligned} F_1 = \overline{A}\overline{B} + \overline{B}C + AC &= m_1 + m_4 + m_5 + m_7 \\ &= \overline{\overline{m_1 + m_4 + m_5 + m_7}} = \overline{\overline{m_1} \cdot \overline{m_4} \cdot \overline{m_5} \cdot \overline{m_7}} \\ &= \overline{\overline{Y_1} \cdot \overline{Y_4} \cdot \overline{Y_5} \cdot \overline{Y_7}} \end{aligned}$$

$$\begin{aligned} F_2 = \overline{A}\overline{B} + \overline{B}C + ABC &= \sum m(0,1,2,6,7) = \overline{\overline{m_0} \cdot \overline{m_1} \cdot \overline{m_2} \cdot \overline{m_6} \cdot \overline{m_7}} \\ &= \overline{\overline{Y_0} \cdot \overline{Y_1} \cdot \overline{Y_2} \cdot \overline{Y_6} \cdot \overline{Y_7}} \end{aligned}$$

$$F_3 = \overline{A}C + BC + A\overline{C} = \sum m(1,3,4,6,7) = \overline{\overline{Y_1} \cdot \overline{Y_3} \cdot \overline{Y_4} \cdot \overline{Y_6} \cdot \overline{Y_7}}$$

4.10 基于基本组合逻辑功能部件的组合逻辑设计



4.10 基于基本组合逻辑功能部件的组合逻辑设计

例2：用2-4译码器和适当的逻辑门实现逻辑函数

$$F_1 = \overline{A}\overline{B} + \overline{B}C + AC$$

$$F_2 = \overline{A}\overline{B} + \overline{B}\overline{C} + ABC$$

$$F_3 = \overline{A}C + BC + A\overline{C}$$

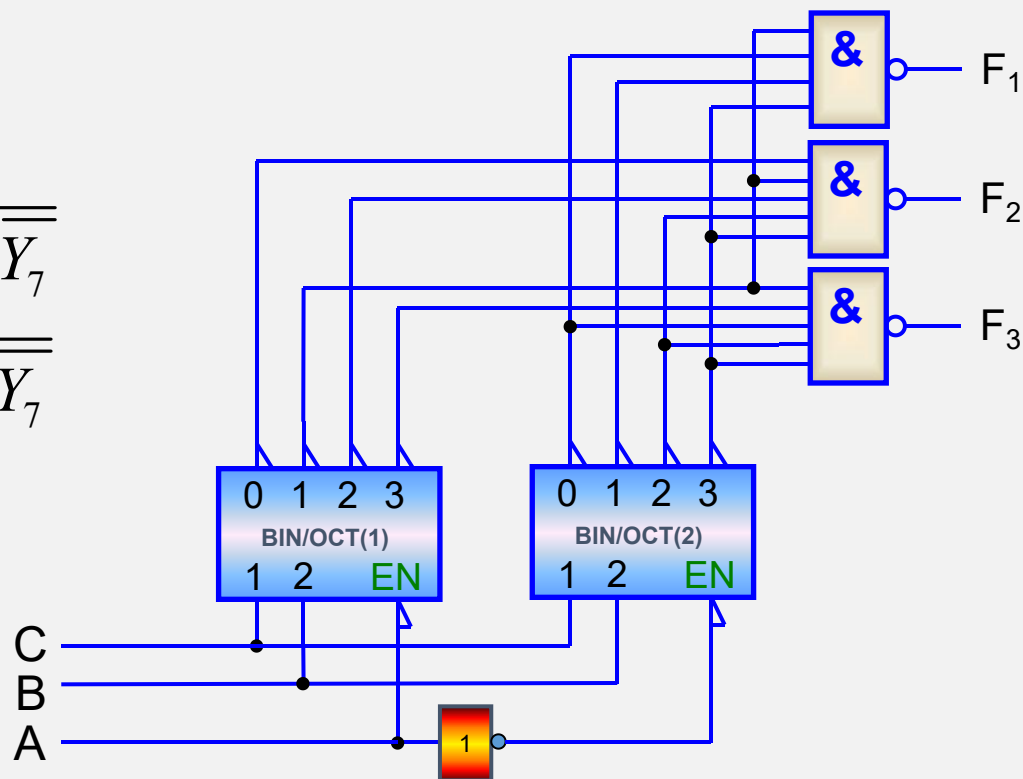
$$F_1 = \overline{A}\overline{B} + \overline{B}C + AC = \overline{\overline{m_1} \cdot \overline{m_4} \cdot \overline{m_5} \cdot \overline{m_7}} = \overline{\overline{Y_1} \cdot \overline{Y_4} \cdot \overline{Y_5} \cdot \overline{Y_7}}$$

$$F_2 = \overline{A}\overline{B} + \overline{B}\overline{C} + ABC = \overline{\overline{m_0} \cdot \overline{m_1} \cdot \overline{m_2} \cdot \overline{m_6} \cdot \overline{m_7}} = \overline{\overline{Y_0} \cdot \overline{Y_1} \cdot \overline{Y_2} \cdot \overline{Y_6} \cdot \overline{Y_7}}$$

$$F_3 = \overline{A}C + BC + A\overline{C} = \overline{\overline{m_1} \cdot \overline{m_3} \cdot \overline{m_4} \cdot \overline{m_6} \cdot \overline{m_7}} = \overline{\overline{Y_1} \cdot \overline{Y_3} \cdot \overline{Y_4} \cdot \overline{Y_6} \cdot \overline{Y_7}}$$

4.10 基于基本组合逻辑功能部件的组合逻辑设计

$$F_1 = \overline{\overline{Y_1} \cdot \overline{Y_4} \cdot \overline{Y_5} \cdot \overline{Y_7}}$$
$$F_2 = \overline{\overline{Y_0} \cdot \overline{Y_1} \cdot \overline{Y_2} \cdot \overline{Y_6} \cdot \overline{Y_7}}$$
$$F_3 = \overline{\overline{Y_1} \cdot \overline{Y_3} \cdot \overline{Y_4} \cdot \overline{Y_6} \cdot \overline{Y_7}}$$



4.10 基于基本组合逻辑功能部件的组合逻辑设计

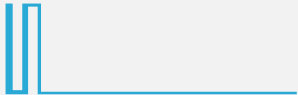


$$Y = \overline{A_1}\overline{A_0}D_0 + \overline{A_1}A_0D_1 + A_1\overline{A_0}D_2 + A_1A_0D_3$$

$$F_1 = \overline{A}\overline{B} + \overline{B}C + AC = m_1 + m_4 + m_5 + m_7$$

$$F_2 = \overline{A}\overline{B} + B\overline{C} + ABC = \sum m(0, 1, 2, 6, 7)$$

$$F_3 = \overline{A}C + BC + A\overline{C} = \sum m(1, 3, 4, 6, 7)$$



本节内容完